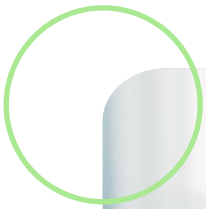




DIPLOMA IN

DATA ANALYSIS

Lesson 3: Probability



Data Analysis Toolpak <

Intro to Probability <

Intro to Random Variables <

Lesson Objectives

Lesson 3



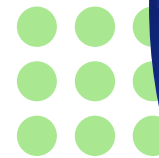
Excel: Data Analysis Toolpak





What is the Data Analysis Toolpak?

- Add-in feature of Excel
Provides data analysis tools
Financial, statistical,
engineering



Why use the Data Analysis Toolpak?

- Saves time
- Easy to use
- Simple analysis



Should you play the lottery?

42 million combinations

The only certainty of this game, is the more tickets you purchase, the higher probability that you will lose





Intro to Probability



Probability?

Used in everyday lives

- Objectively
- Subjectively





Why probability?

Creates likelihoods in an uncertain world

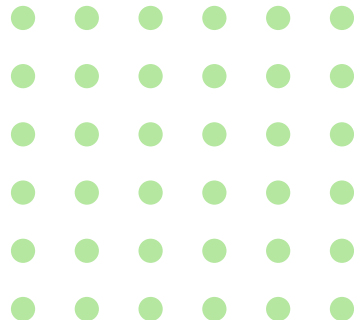
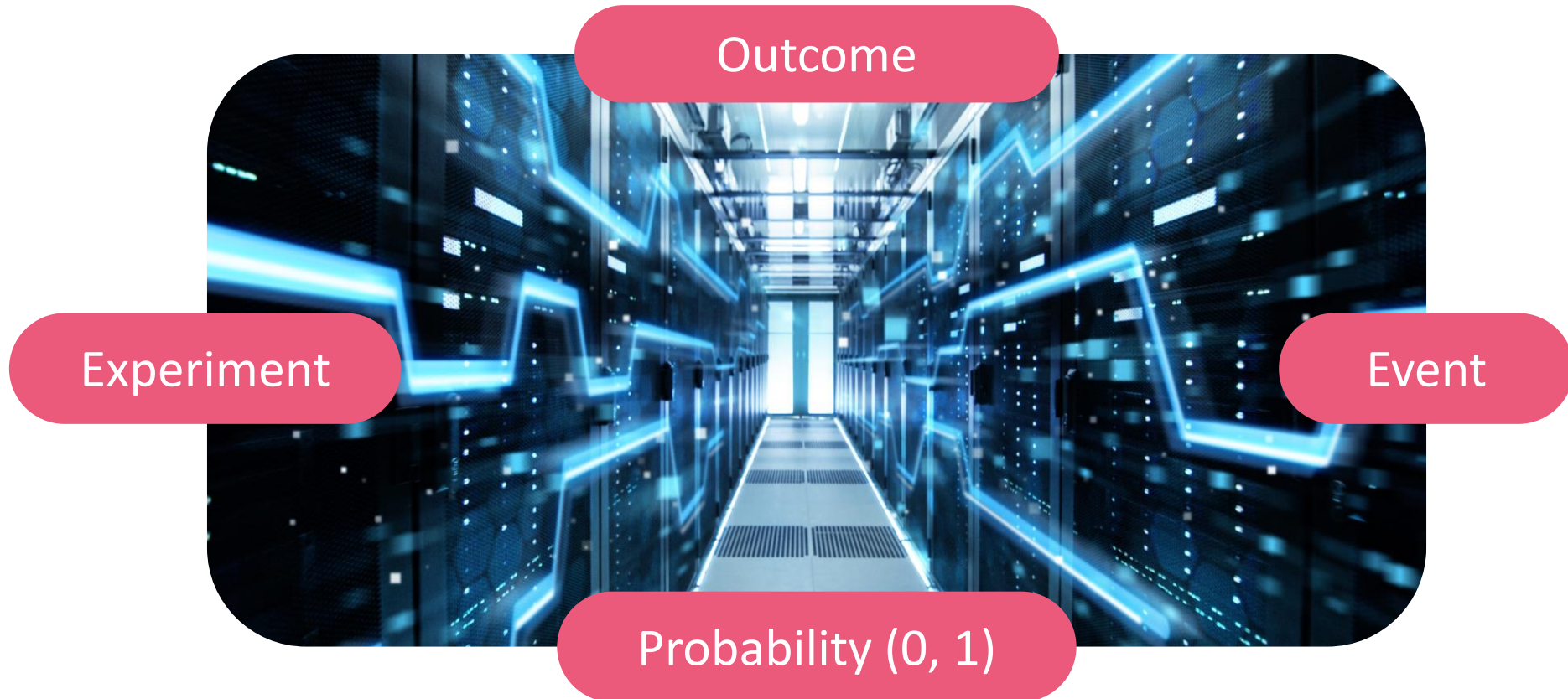
Backbone of data analysis

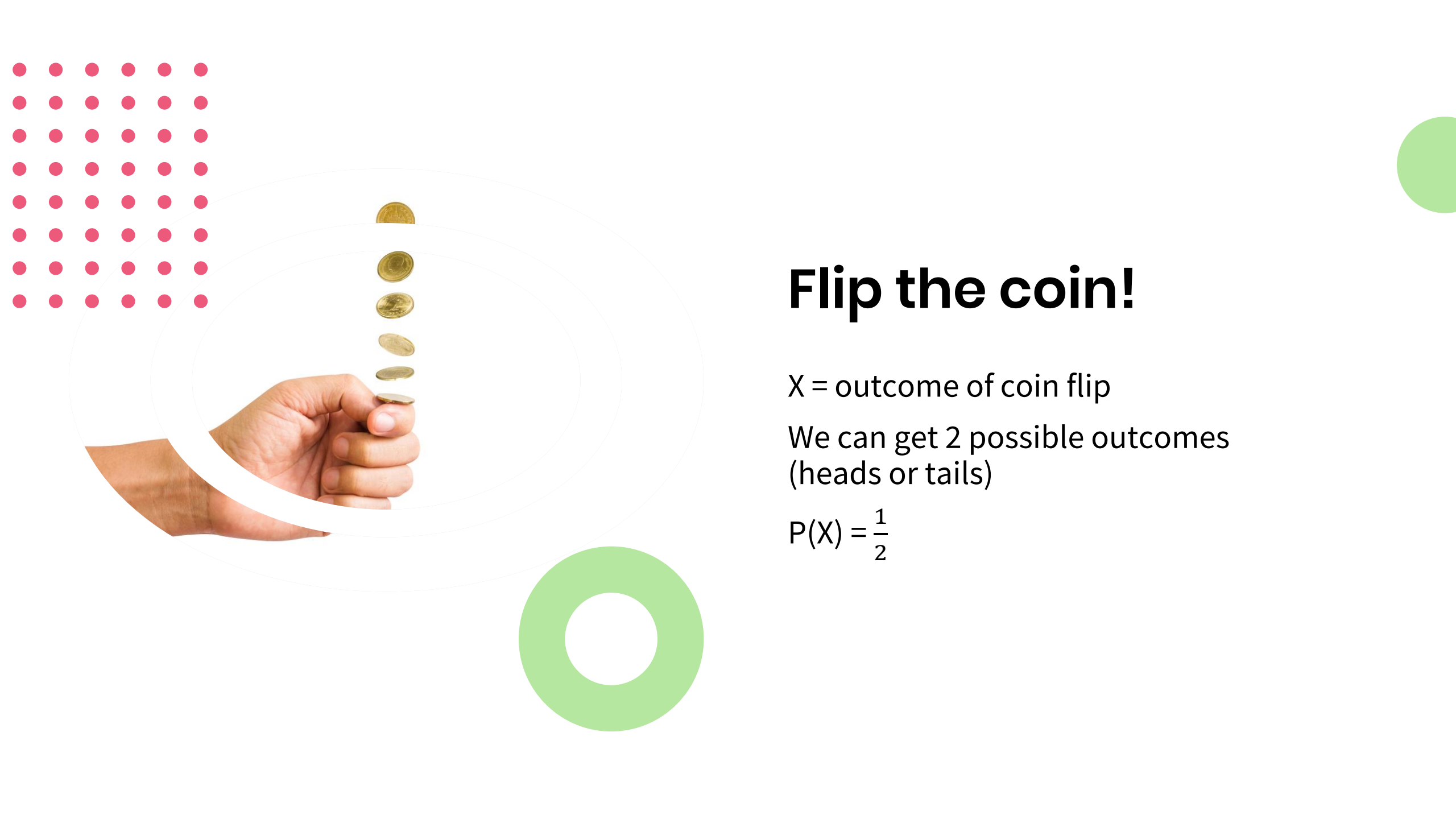
Helps to know, understand and prepare better for chances of events

- E.g. Is it likely to rain? Am I more likely to develop diabetes because of my diet?

Helps make statements about how much confidence we have in estimates (means, proportions, etc.)

➤➤ Probability terms





Flip the coin!

X = outcome of coin flip

We can get 2 possible outcomes
(heads or tails)

$$P(X) = \frac{1}{2}$$

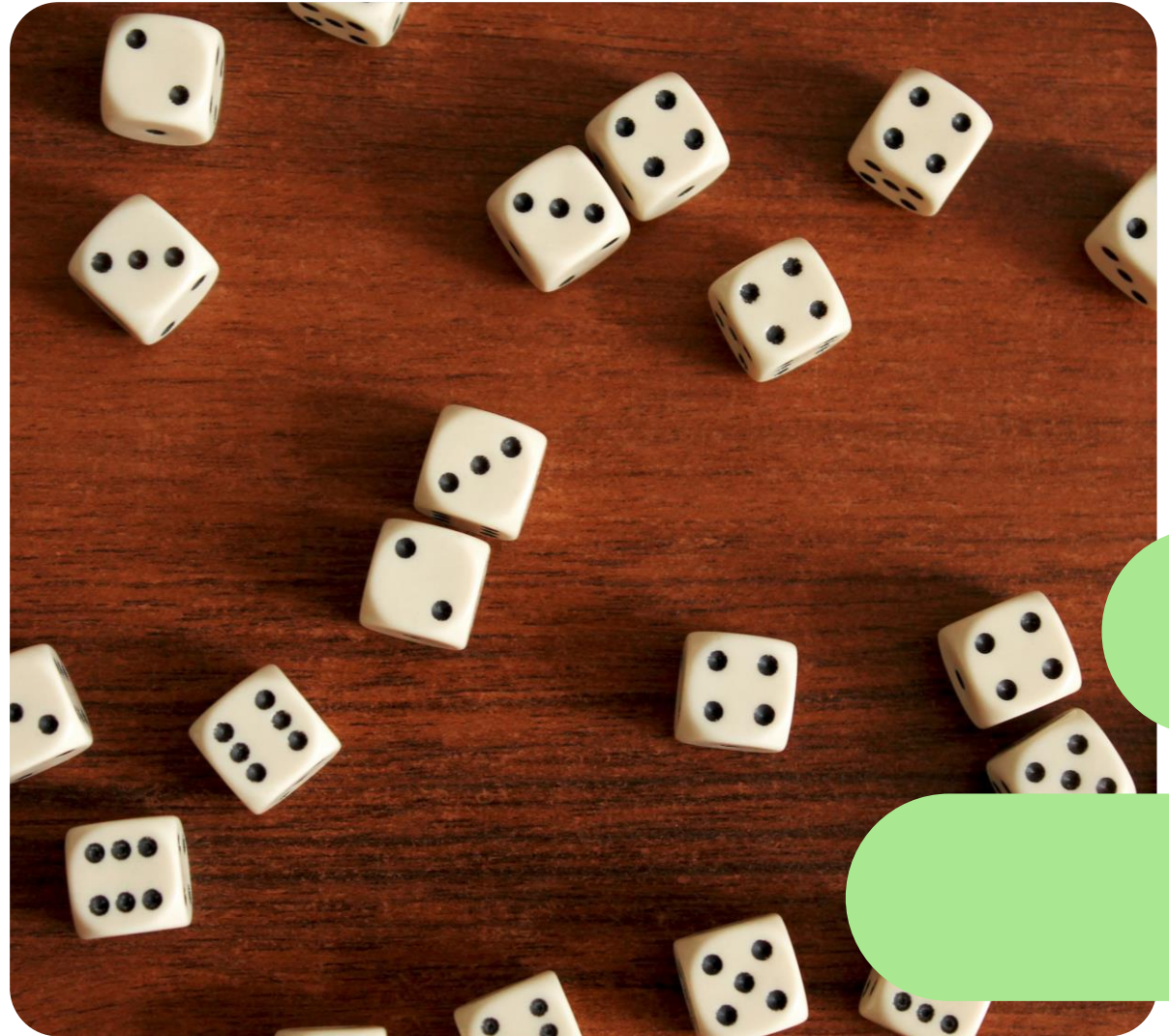


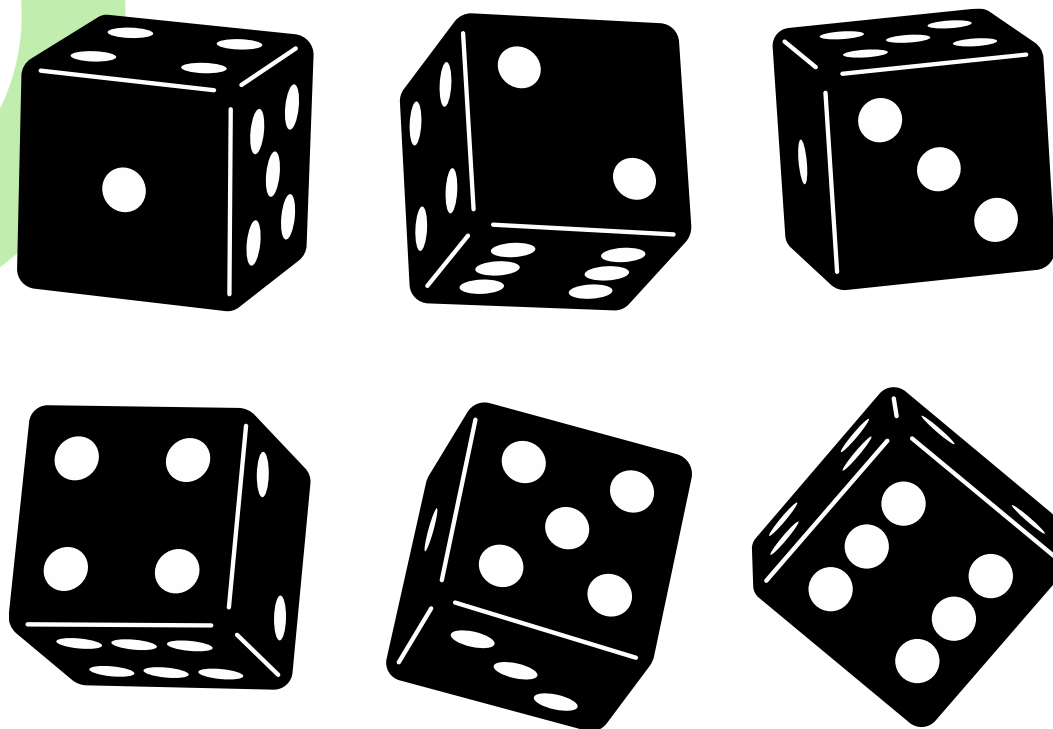
Flip the coin 10 times

- No guarantee for 5 tails
- As number of flips increases, probability of observing tails tends to $\frac{1}{2}$
- $P(X) = \frac{3}{6} = \frac{1}{2}$

Roll the dice!

- To win, I need to get sum of 8 with 2 dice rolled
- Let Y be sum of two dice number's that face upwards
- What is probability of winning?





Sample space

(Space that contains all the possible outcomes of the experiment)

Ω

- X = sum of 2 throws
- $X = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$

E.g.

$$2 = 1 + 1$$

$$3 = 2 + 1$$

$$4 = 2 + 2$$

Probability of each number

Total combinations = $36 = (6 \times 6)$

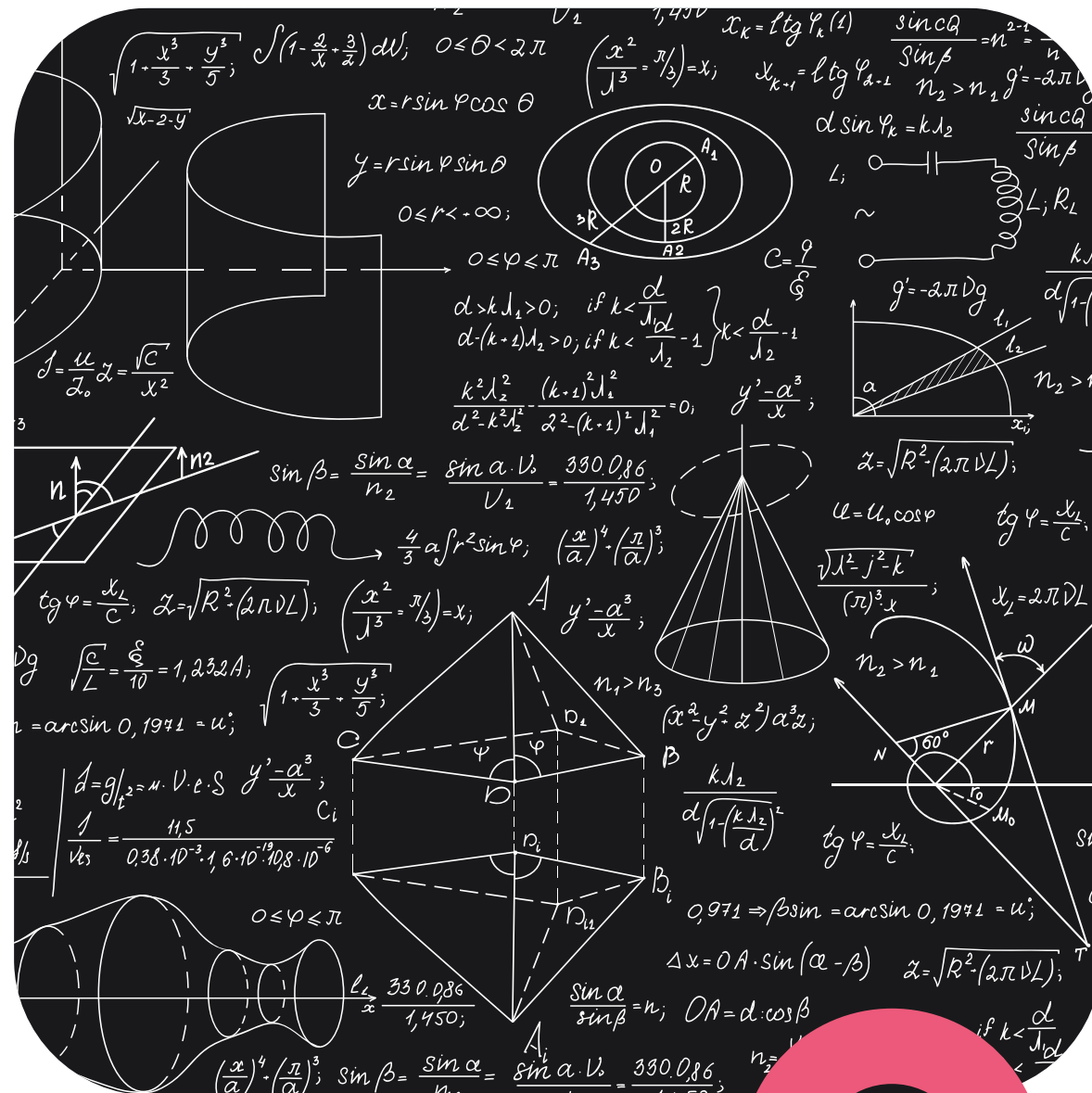
- $2\{(1,1)\} : \frac{1}{36}$
- $3\{(1,2), (2,1)\} : \frac{2}{36}$
- $4\{(2,2), (3,1), (1,3)\} : \frac{3}{36}$
- ...
- $8\{(2,6), (6,2), (3,5), (5,3), (4,4)\} : \frac{5}{36}$

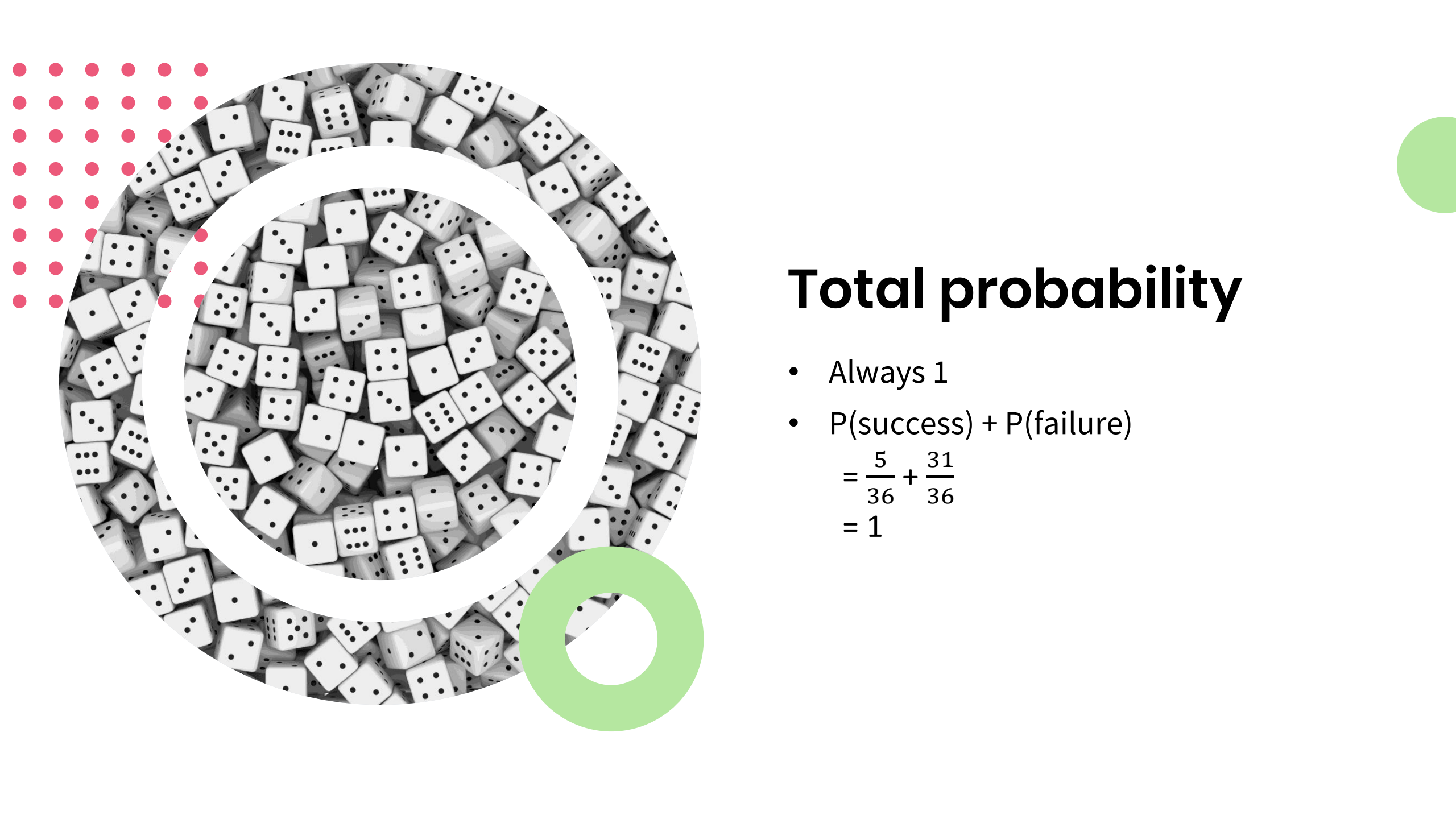


Chances of success for sum of 8?

- P(sum of 8 from 2 dice throw)
= P(success)
$$= \frac{5}{36} = \frac{1}{7}$$
- P(not throwing sum of 8)
= P(failure)
= 1 – P(success)
$$= 1 - \frac{5}{36}$$

$$= \frac{31}{36}$$





Total probability

- Always 1
- $P(\text{success}) + P(\text{failure})$
 $= \frac{5}{36} + \frac{31}{36}$
 $= 1$



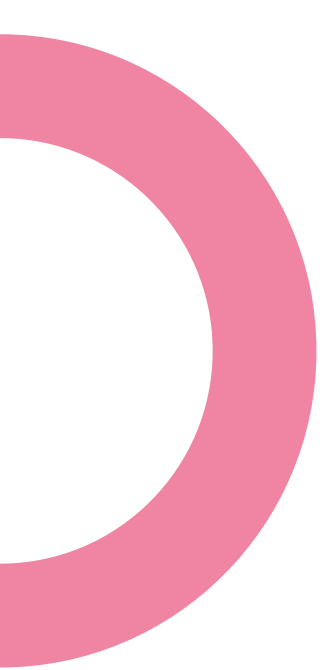
so Random



What is a random variable?

- A variables with values that are numerical outcomes of random experiment/study
- Function mapping outcomes of random process to numerical value



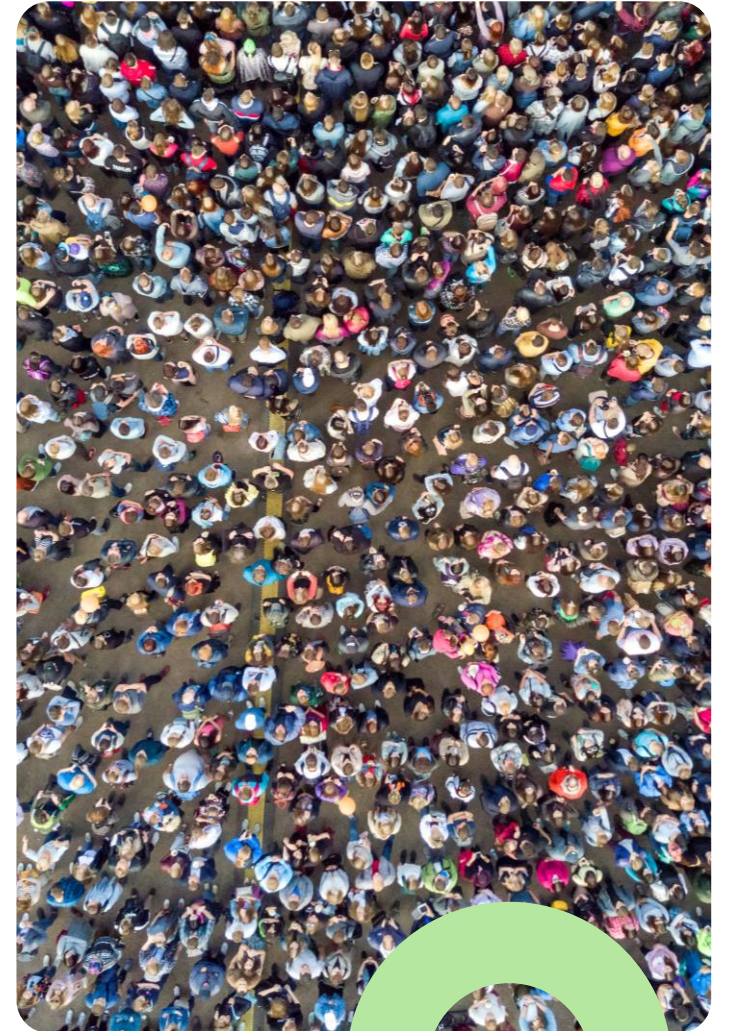


**Random
variables
can be**

Discrete



Continuous





Revisiting our coin flip example

- Random variable X = outcomes of coin flip
- $X = 1$ (coin flip lands on head)
- $X = 2$ (coin flip lands on tail)



Revisiting our dice example

Random variable Y = sum of
upwards facing dice numbers



Challenge »

Install Data Analysis
Toolpak in Excel and use
Descriptive Statistics feature
on Titanic data

#exploredata